SIMATS ENGINEERING
ASSESSMENT TOOL 3: Application Based Questions

COURSE NAME: Cloud Computing and Big Data Analytics **COURSE CODE:** CSA1520

COURSE OUTCOME COVERED
CO2: Analyze virtualization architectures to identify performance and security issues in cloud computing environments. (BL3)

Dave's Taxonomy: Manipulation (Level 2) Total Marks: 25
Question Count: 5

Code of Conduct
I Kurisha Judith J (192572071) (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.
I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student: 

Assessment Tasks

Application Based Questions

1. A company experiences unpredictable traffic spikes during seasonal sales. Explain how cloud auto-scaling and load balancing can handle this demand efficiently. How does this approach reduce downtime and improve user experience? What role do monitoring tools play in this setup? Relate your answer to a real-world e-commerce scenario.

2. An organization wants to store large volumes of backup data securely in the cloud.
   Discuss how cloud storage solutions ensure durability and data redundancy.
   How can encryption and access control protect sensitive backup data?
   Explain the cost advantages compared to traditional storage systems.
   Provide an example of a disaster recovery use case.

3. A mobile app company plans to deploy its application globally. Explain how content delivery networks (CDNs) improve performance and reduce latency. How do edge servers enhance user experience across regions? Discuss how cloud providers support global availability. Relate this to a real-time streaming or gaming application.

4. A financial institution requires strict compliance and data security in cloud usage. Explain how cloud providers support regulatory compliance and auditing. What role do logging, monitoring, and encryption play in security? How can the company ensure data sovereignty requirements are met? Give an example involving sensitive financial transactions.
5. A development team wants to adopt containerization for application deployment. Explain how containers differ from virtual machines in terms of performance and resource usage. How does orchestration (like Kubernetes) simplify deployment and scaling? Discuss portability benefits across different cloud environments. Provide an example of a microservices-based application.

Application Based Questions Rubric

Criteria	**Weightage**	**Level 4 - Exemplary**	**Level 3 - Proficient**	**Level 2 - Developing**	**Level 1 - Beginning**
Relevance to Application	25%	Response directly addresses the given use case with strong relevance	Mostly relevant response	Partially relevant	Weak relevance
Use of Cloud Concepts	25%	Applies appropriate concepts accurately	Mostly accurate application	Basic application only	Incorrect concept usage
Practical Insight	20%	Shows strong real-world understanding	Shows reasonable practical understanding	Limited practical insight	No practical insight
Completeness	15%	Response covers all required aspects	Covers most aspects	Covers only basic aspects	Incomplete response
Clarity	15%	Clear and concise answer	Generally clear	Some confusion in expression	Poor clarity

1) Cloud auto-scaling and Load Balancing during seasonal sales

Cloud auto-scaling and load balancing help companies manage unpredictable traffic spikes during seasonal sales efficiently. For example, an e-commerce website may receive a large number of visitors during Diwali or Black Friday sales.

Auto scaling:

It is a cloud computing feature that automatically increases or decreases computing resources according to demand. When the number of users increases, the cloud platform launches additional virtual machines or application instances. When traffic decreases, unnecessary resources are removed. This ensures sufficient resources are available while reducing operational costs.

Load balancing:

It distributes incoming user requests across multiple servers. Instead of allowing one server to handle all requests, a load balancer directs traffic to available servers based on their workload and health. This prevents server overload and improves application

performance.
Together, these technologies reduce downtime by maintaining sufficient server capacity and redirecting requests away from unhealthy servers. users can browse products, add items to their carts, and complete payments with fewer delays, improving their overall experience.

Monitoring tools:
It play an important role in this setup. They continuously track CPU utilization, memory usage, response time, request rates, and server health. When resource usage exceeds predefined thresholds, monitoring alerts or auto-scaling policies can trigger additional resources. Examples include ~~Azzo~~ Amazon cloudWatch and Azure monitor.

Real-world example: During a major online shopping sale, an e-commerce company may experience a sudden increase in visitors. Auto-scaling adds more application instances, while the load balancer distributes customer requests among them. Monitoring tools identify performance problems and help administrators

respond quickly.
Thus auto-scaling, load balancing and monitoring work together to improve availability, maintain performance, reduce downtime and provide a smooth shopping experience during peak traffic.

2) Secure cloud Backup storage and Disaster Recovery

cloud storage enables organizations to store large volumes of backup data securely, reliably and cost-effectively. cloud providers offer storage services designed to protect data against hardware failures, accidental deletion and disasters.

Data durability and redundancy:
cloud storage providers maintain multiple copies of data or use redundancy mechanisms such as replication and erasure coding. Data may be stored across different devices or availability zones. If one storage device fails, the data can still be recovered from other copies. Versioning and immutable backups can also help protect against accidental deletion and ransomware attacks.

Encryption and access control:

Encryption protects sensitive backup data from unauthorized access. Encryption at rest protects stored data, while encryption in transit protects data transferred between systems and cloud storage. organizations can use encryption keys managed by the cloud provider or manage their own keys.

Access control mechanisms, such as identity and Access Management (IAM), ensure that only authorized users and applications can access backup data. Multi-factor authentication, least-privilege permissions and audit logs provide additional security.

cost advantages:

Traditional storage systems require organizations to purchase hardware, maintain data centers, provide electricity and employ maintenance staff. cloud storage reduces these initial investments because organizations generally pay for the storage and services they use. Different storage classes are available for frequently accessed data and long-term archives, helping reduce costs.

Disaster recovery example:

suppose a company's local data center becomes unavailable due to a flood or fire. If its databases and important files have been backed up to a separate cloud region, the company can restore the data and restart its applications in another location. This reduces data loss and service interruption.

Therefore, cloud storage provides durability, redundancy, security, scalability and cost savings. Regular backup testing and a well-planned disaster recovery strategy are essential to ensure that data can be restored successfully when required.

3) Content Delivery Networks (CDNs) and global Application Deployment.

Content Delivery Networks (CDNs) help mobile applications deliver content quickly to users across different geographical regions. A CDN is a distributed network of servers that stores cached copies of content, such as images, videos, application files and other static resources, closer to users.

Improved performance and reduced latency:

When a user requests content, the CDN directs the request to a suitable nearby server. since the content travels a shortest distance, loading time and network latency are reduced CDNs also reduce the workload on the main origin server and help manage large volumes of traffic.

Role of edge servers:

Edge servers are located close to end users and deliver cached content quickly. some edge platforms can also execute lightweight application code near users. This reduces the need to communicate with a distant central server for every request and improves responsiveness across different regions.

Global availability:

cloud providers operates data centers and availability zones in multiple geographical regions. They support global applications through regional deployments, traffic routing, health monitoring, replication and failover mechanisms. If one server or region becomes unavailable, traffic can be redirected to another healthy

location, provided the application is configured for regional recovery.

Real-world example:

Consider a video streaming application such as Netflix. When a user watches a movie, video content can be delivered from a nearby CDN server rather than repeatedly being transferred from a distant central server. This reduces buffering, improves playback quality, and supports many users simultaneously. Similarly, online gaming applications use geographically distributed servers to reduce communication delays and provide a smoother gaming experience.

Thus CDNs, edge servers and global cloud infrastructure improve application speed, reduce latency, distribute traffic efficiently and increase availability for users worldwide.

4) Cloud security, Regulatory compliance and Data Sovereignity in Financial Institutions

Financial institutions handle sensitive information such as customer details, bank account information, transaction records,

and payment data. Therefore, they must ensure strong security, regulatory compliance, and proper control over where data is stored and processed.

Regulatory compliance and auditing:

cloud providers offer security features and compliance documentation that help organizations meet applicable regulatory requirements. Depending on the service and scope, providers may support standards such as ISO 27001 and PCI DSS. However, however the financial institution remains responsible for configuring its cloud environment correctly and complying with applicable laws and regulations.

Auditing helps verify whether security policies are followed. Regular audits examine user permissions, data access, system configurations and security controls. Audit reports provide evidence for internal reviews and regulatory inspections.

Role of logging and monitoring:

Logging records important activities, including login attempts, account changes, and access to sensitive data. Monitoring tools analyze these records to identify unusual activity, suspicious transactions, and possible security threats. Alerts help administrators investigate incidents and respond quickly.

Encryption and access control:

Encryption protects data both at rest and in transit. Identity and Access Management (IAM), multi-factor authentication, and least privilege permissions ensure that only authorized employees and applications can access sensitive information. Encryption key management provides additional protection.

Data sovereignty:

Data sovereignty refers to the legal requirements associated with the country or jurisdiction where data is stored or processed. A financial institution can select appropriate cloud regions, apply location restrictions and review backup and disaster recovery locations to satisfy applicable requirements.

Example:

When a customer transfers money through a banking application, the institution encrypts transaction data, verifies the user's identity, records the transaction in audit logs, and monitors for suspicious activity. customer records are stored in approved geographical locations according to applicable regulations.

Therefore, security controls, auditing, monitoring, encryption, and careful region selection help financial institutions protect customer information and maintain compliance when using cloud services.

5) containerization, virtual machines, and orchestration

containerization is a technology that packages an application and its required libraries, dependencies, and configuration into a container. containers allow applications to run consistently across development, testing and production environments.

Containers versus virtual machines:

Virtual machines run on a hypervisor and usually contain a complete guest operating system along with the application. Containers share the host operating system's kernel while keeping applications isolated from one another. Containers generally start faster, require fewer resources and allow more application instances to run on the same hardware. Virtual machines provide stronger operating system-level isolation and can run different operating systems on the same physical machine. Both technologies are useful, depending on security and workload requirements.

Role of orchestration:

Container orchestration tools such as Kubernetes automate the deployment, management, and scaling of containers. They schedule containers across available machines, restart failed containers, perform health checks and distribute traffic. Kubernetes can also increase or decrease application replicas according to demand.